

Distractive or supportive – how warnings in the head-up display affect drivers' gaze and driving behavior

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Abstract— Urban areas expose drivers to a lot of challenges (e.g., many kinds of distracting stimuli and road users) and critical situations, increasing the probability of driver errors. Therefore the accident rate in urban areas is quite high and driver assistance is needed. As part of the research project UR:BAN, this study investigates the effectiveness of four different types of driver warnings (+ control group), presented in a head-up display (HUD), to support drivers in a very critical urban scenario. The aim of these driver warnings was to alert drivers to a forthcoming pedestrian crossing the ego vehicle's road, eliciting a fast and strong response from the drivers like an emergency brake. The driver warnings varied in their strategy (attention vs. reaction oriented) and specificity (generic vs. specific). In a driving simulation the drivers' gaze and driving behavior were analyzed. A total of sixty drivers were tested in a between-subjects design (27 female, 33 male; $M = 23.7$ years, $SD = 3.7$ years). In general, all driver warnings affected the drivers' performance positively. Even though the number of collisions was not reduced, drivers showed a faster and stronger brake reaction when being warned, which nevertheless reduced the collision severity. While all drivers gazed at the safety-critical object, only about half of the drivers showed a HUD glance. When the drivers gazed at the HUD, the positive effect of the driver warnings on the brake reaction time was reduced. Thus, driver warnings which do not have to be fixated in order to be perceived are an interesting perspective for future research. Furthermore, the warning types differed little from each other in their effects on the drivers' performance. However, one warning, which was somewhat more unusual and difficult to understand, was least effective. This shows that a bad design of a warning can negate the potential benefits. Overall, the examined driver warnings can be recommended for very critical situations as they improved the drivers' performance and did not distract them. However, not all driver warnings are equally well suited.

Keywords— driver warning; accident prevention; driving and gaze behavior; head-up display; urban area

I. INTRODUCTION

Urban areas and their complex driving situations are a big challenge to drivers. They are confronted with numerous distracting stimuli and high information density, combined with short decision times. Additionally, the interaction between various road users (vehicles, pedestrians, bicyclists) is at its maximum here. Thus, many critical situations arise in urban areas, combined with driver errors this result in a large number of accidents. Assisting drivers here is of great importance and

thus is also aim of the research project UR:BAN. The following study, as being part of this project, investigates the effectiveness of different visual driver warning concepts in supporting drivers in critical urban situations.

Recent research concentrates on developing methods to detect objects (e.g., pedestrians, bicyclists, obstacles) in the driving environment and assess the hazard they present in order to trigger driver (collision) warnings if needed [1]. Aim of these is to alert drivers to the impending critical situation so that they can react adequately such as by braking or swerving. For example, in a very critical situation with imminent collision risk, driver warnings should elicit a fast and strong brake reaction from drivers that could reduce the occurrence and severity of accidents. In order to achieve this, the design of driver warnings becomes very important.

First of all, driver warnings have to be perceived by drivers. For that purpose they should be salient enough to attract drivers' glances even in time-critical situations, but without distracting the drivers from the road and the critical situation. This brings the head-up display (HUD) into focus as a location for driver warnings. As compared to head-down displays (HDD) like instrument panels [2] [3] or center column displays [4], the HUD projects warnings into the windshield. Thus, drivers can perceive them in their field of view, overlapping the road reality, which means shorter eye accommodation times and shorter gaze retention times than instrument panels [5] or center column displays [4]. Furthermore, displaying driver warnings in the HUD lessens the chances of overseeing a critical event in the environment and saves time for the drivers to react upon such [6].

However, driver warnings should not capture the drivers' glance for too long, which might occur if drivers do not immediately understand the warnings, perhaps causing them to be distracted from the actual critical situation and slowing down their reaction. Hence, driver warnings should be designed in a way that drivers can perceive them promptly, understand the enclosed information intuitively and process it rapidly [7] in order to assess the critical situation appropriately. This is supported by [8], who found an increase in driver reaction and detection times upon changes with increasing cognitive demand of warnings. As traffic signs are well known to all drivers from driver license training and everyday traffic, their use as driver warnings might help drivers to process and

understand critical situations better, supporting them optimally to select and trigger an adequate reaction [9] [10] [11] [12].

In very critical scenarios the time left to prevent a collision with a safety-critical object is limited. During this short time one might concentrate primarily on raising and/or guiding the drivers' attention in order to improve the understanding of the critical situation (attention oriented), while another warning strategy might skip this step and try to elicit automatic driver responses (reaction oriented). Furthermore, the specificity of the driver warnings may matter. Being specific, they might provide drivers with additional details about the critical situation like the safety-critical object involved (e.g., pedestrian or bicyclist) or the optimal reaction choice (e.g., braking or steering). In contrast, driver warnings could also be generic to avoid the presentation of information which might take drivers too much time to process and thus slow down their reaction. Each warning strategy (attention vs. reaction oriented) in its specificity (generic vs. specific) has advantages and disadvantages. For example, generic driver warnings may lack important details for the understanding of the situation leading to confusion and thus increasing reaction times or they might immediately trigger an automatic driver reaction like emergency braking and thus save time. Aim of this paper is to analyze which warning type would be most effective in supporting drivers in critical urban situations.

Exemplary for a very critical urban situation a pedestrian crossing scenario is used in this study, as this type of accident leads to the most deaths in urban areas (26.8% of all fatal accidents in German urban areas in 2013 [13]). The scenario only leaves enough time for a late warning timing, additionally contributing to its high criticality. In this situation driver distraction through the warnings (e.g., by guiding the drivers' glance away from the relevant scene) might have the most negative impact. If the proposed driver warnings show positive effects over no warning even in this highly critical situation, they could be recommended in general. Therefore this scenario is analyzed in detail in this paper, while the other seven scenarios, examined for drivers to experience the warning in diverse situations, will be reported in a further paper.

To summarize, in this driving simulator study the effects of four different driver warnings in the HUD were analyzed in a pedestrian crossing scenario. The aim of this paper is to give a detailed analysis of the drivers' performance after warning presentation. Special interest lies on how the different warning types influence the attention processes measured by eye-tracking and how that affects the brake reactions. Positive effects of every warning type over no warning were expected, as well as differences in the driving and gaze behavior due the warning type (varied in warning strategy and specificity).

II. METHODS

A. Driver warnings

Drivers were supported by visual warnings in form of traffic signs displayed in a multicolor HUD. As any additional acoustic or haptic warning signal might have masked the visual effects of the different warning types, the driver warnings were exclusively given visually. They were projected into the

windshield overlapping with the scenery. Presented driver-centered, right above the engine hood, measuring maximum 21x21 cm (6° visual angle), they were triggered as the safety-critical object started moving (around 1.66 s before its crossing point with the ego vehicle). Moreover, the driver warnings varied in their strategy (reaction vs. attention oriented) and specificity (generic vs. specific). In sum, four different warning types were generated (see Table I). As reaction oriented generic (RG) warning, a stop sign was to elicit an immediate brake reaction by the drivers, while the reaction oriented specific (RS) warning displayed a stop sign for brake reactions or a swerving sign when steering reactions were more beneficial like in the analyzed scenario. In the attention oriented generic (AG) warning the drivers' general attention was to be raised by a caution sign, whereas in the attention specific (AS) warning the triangle included the current safety-critical object (a pedestrian in the analyzed scenario) in order to guide the drivers' attention directly towards it. Table I shows all implemented warning types as well as the control group (C), receiving no warning at all.

TABLE I. IMPLEMENTED WARNING VISUALIZATIONS IN THE HUD WITH THE WARNING TYPES PRESENTED IN THE VERY CRITICAL SCENARIO IN COLOR

Warning type	Presentation in head-up display
1) Control (C)	No warning
2) Reaction generic (RG)	
3) Reaction specific (RS)	 
4) Attention generic (AG)	
5) Attention specific (AS)	   

B. Scenario and simulator

The study was conducted in the fixed base medium fidelity driving simulator of the Technische Universität Braunschweig. It consists of a seat box with typical car interiors like a steering wheel with force feedback, accelerator and brake pedals as well as two LCD screens serving as rear-view mirrors. The virtual urban scenery created by the driving simulation software SILAB 3.0 (from WIVW [14]; see www.wivw.de) is projected onto three silver screens (left, ahead, right), providing the drivers with a 180° field of view at about 2.1 m distance from the driver's seat. In the examined very critical driving scenario the ego vehicle was driving straight ahead with 50 km/h on a one-way road with a free green area to the left and a long row of parking cars to the right, when suddenly a pedestrian crossed the ego vehicle's path from right to left. The pedestrian was occluded from sight by a small van before starting to cross. It was activated to a speed of 2 m/s as soon as the drivers passed a fixed point (hedgehog) in the simulated scene, being around 23 m before the pedestrian crossing point. Drivers had to react immediately, either by very strong braking or swerving onto the green area in order to avoid an imminent collision with the pedestrian. Although both reactions would have been efficient, all drivers braked, whereas almost no one swerved. So the analysis will focus on the braking behavior.

C. Participants

In total, sixty drivers (27 female, 33 male) participated in the experiment (Mean (M) = 23.7 years, Standard Deviation (SD) = 3.7 years). They were recruited through flyers, mailing lists and an existing pool of trained simulator drivers. The average driving experience was 5.7 years (SD = 3.6 years) and the annual mileage was mainly less than 3000 km. All participants had normal or corrected-to-normal visual acuity. Simulator training was required for participation in order to avoid simulation sickness. Drivers were compensated with 12 € for a successful participation or received course credits (if students at the Technische Universität Braunschweig).

D. Procedure

After a welcome, participants were instructed in written form about the experiment's objectives. They signed a consent form and filled in a demographic questionnaire. To familiarize the drivers with the simulator, they absolved an about 15 min training drive in the simulator. Subsequently equipped with the Dikablis glance tracking glasses, they started the actual test drive with in sum eight different critical situations. After the first test drive, a small break was offered, before the second test drive started. Both test drives included the same scenarios, however in randomized orders. In the end, the participants were interviewed about the driver warnings and their usability and thanked for their participation as well as compensated. The entire experiment took approximately one and a half hour.

E. Data and data analysis

For the experiment, a between-subjects design was used. Each driver experienced one warning type in the critical situation. Accordingly, there were five groups, a control group without a warning and four warning groups receiving different warning types (warning types: no warning (C), reaction oriented generic (RG), reaction oriented specific (RS), attention oriented generic (AG), attention oriented specific (AS), see Table I). The paper analyzes gaze (Dikablis) and driving behavior data (SILAB) separately and in combination focusing on the results of the first test drive. Table II gives an overview of the measured variables analyzed in this paper.

TABLE II. ANALYZED VARIABLES

Variable	Unit	Description of variable
AOI glance	-	Number of drivers who gazed at an area of interest (AOI: safety-critical object or HUD) for at least 0.1 s
AOI glance duration	s	Time period in which an AOI (safety-critical object or HUD) is fixated
AOI glance reaction time	s	Time from warning onset until AOI (safety-critical object or HUD) fixation
Number of collisions	-	Number of incidents, when the ego vehicle collided with the crossing pedestrian
Brake reaction time	s	Time from warning onset until pressing of the brake pedal
Maximum braking value	%	Maximum pressing of the brake pedal after warning onset in percent of the sample maximum before pedestrian crossing point

For inferential statistical data analysis IBM SPSS Statistics 22 was used. When applicable a general linear model with the

between-subjects factor warning type or a t-test with the factor HUD glance was conducted. For the number of collisions a chi-squared test was computed. A significance level of alpha $p = .05$ was adopted for all statistical tests. In all other cases descriptive analyses are reported, especially for all data concerning the HUD glance due to a reduced number of participants (N), who actually gazed at the HUD. This will be explained in detail in the following section.

III. RESULTS

A. Gaze behavior

The analysis of the gaze behavior showed that all drivers sooner or later gazed at the safety-critical object (CO). The HUD however, was gazed at before brake initiation by only 45.8% of all drivers who received a warning (N = 22). The HUD glance distribution over the warning groups was quite equal, with about half of the drivers within each warning group gazing at the HUD (HUD glance before brake initiation for RG: N = 6, RS: N = 5, AG: N = 5, AS: N = 6). Some more HUD glances occurred after drivers began to brake, especially in the RG warning (HUD glance after brake initiation for RG: N = 5, RS: N = 1, AG: N = 1, AS: N = 1). However, in this paper the HUD glances before brake initiation are focused as this is the critical phase with the most distraction potential of the driver warnings.

Inferential statistical analyses of the CO glance showed no significant main effects of the warning type, neither for the CO glance reaction time ($F_{(4,55)} = 0.32, p = .867$) nor the CO glance duration ($F_{(4,55)} = 1.46, p = .228$). Overall, the mean CO glance reaction time was about 0.36 s (SD = 0.28 s) and the mean glance duration about 1.72 s (SD = 0.91 s). However, when directly comparing the single warning types with the control group, the RG warning had a noticeably lower CO glance duration compared to the control group (C-RG: $p = .022$) and in sum the lowest of all warning groups (C: M = 2.10 s, SD = 1.04 s; RG: M = 1.23 s, SD = 0.74 s).

The HUD glances were only analyzed in the four warning groups since no driver warnings were presented in the HUD for the control group. As described above, only five to six drivers within every warning group gazed at the HUD at all. Due to the thereby limited number of participants, statistical analyses became critical. Overall, it took drivers on average about 0.18 s (SD = 0.11 s) to gaze at the HUD. Additionally, these glances lasted on average about 0.24 s (SD = 0.12 s). Regarding the means per warning type, the groups were quite similar except for the RS warning. It led to the shortest HUD glance reaction times (M = 0.04 s, SD = 0.07 s), with the glances of three drivers resting already coincidentally on the HUD before the warning was visible. Additionally, the HUD glances upon the RS warning lasted the longest (M = 0.35 s, SD = 0.10 s).

B. Number of collisions

To analyze the effect of the warning type on the number of collisions, a chi-squared (X^2) test was conducted. As can be seen in Fig. 1, there were no overall significant differences in the number of collisions due to the warning type ($X^2 = 1.60$,

$p = .808$). About half of the drivers encountered an accident in total ($n = 31$) as well as within every group (see Fig. 1).

Fig. 1 further shows the effect of the HUD glance on the number of collisions. In another chi-squared test over all warning types the numbers of collisions did not differ significantly due to the HUD glance either ($X^2 = 3.02$, $p = .082$; share of collisions with HUD glance for RG: 80.0%; RS: 57.1%; AG: 42.9%; AS: 60%). Furthermore, Fig. 1 shows the effect of the warning type on the number of collisions for drivers who showed a HUD glance and for those who did not. The absolute number of collisions with HUD glance as well as its relation to the absolute number of collisions avoided with HUD glance was relatively similar between the warning groups (see Fig. 1). Thus, no general warning type effect on the number of collisions depending on the HUD glance is noticeable.

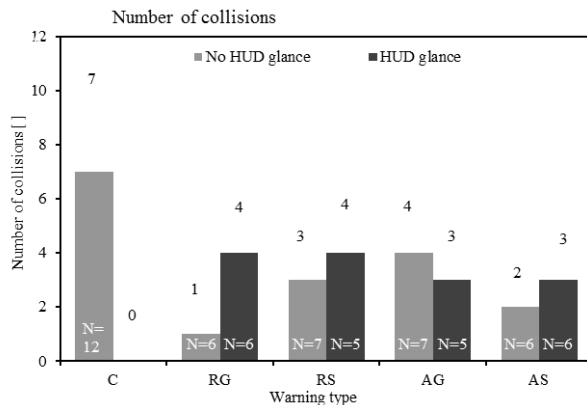


Fig. 1. Number of collisions for the control and all four warning groups with and without head-up display (HUD) glance, displaying the number of participants (N) included in the bottom of the bars.

C. Brake reaction time (Starting to brake)

In order to investigate the influence of the warning type on the brake reaction time, a general linear model with the factor warning type was conducted. It showed a significant overall effect of the warning type on the brake reaction time ($F_{(4,55)} = 2.54$, $p = .050$). Driving with a warning improved the brake reaction time compared to no warning (see Fig. 2), reaching some significant time improvements in post-hoc tests. The RG and AS warning both led to significantly faster brake reactions than the control group (C-RG: $p = .019$; C-AS: $p = .004$). For the AG warning, there was a tendency of an effect (C-AG: $p = .052$), while the RS warning did not differ significantly from the control group (C-RS: $p = .139$).

Furthermore, the t-test over all warning groups showed a significant main effect of the HUD glance on the brake reaction time ($t_{(46)} = -2.12$, $p = .039$). As can be seen in Fig. 2, independent of the warning type drivers gazing at the HUD reacted slower ($M = 0.90$, $SD = 0.12$) than drivers disregarding the HUD ($M = 0.82$ s, $SD = 0.14$). Yet, only in the RS warning group the positive effect of the driver warnings improving the brake reaction time was evened out, when drivers gazed at the HUD. However, the HUD glance still did not delay the brake reaction compared to the control group either.

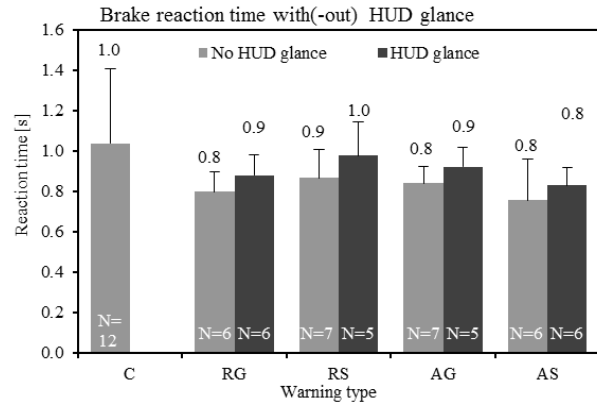


Fig. 2. Mean brake reaction time for the control and all four warning groups with and without head-up display (HUD) glance, displaying the number of participants (N) included in the bottom of the bars.

D. Maximum braking value (Braking process)

Besides reacting fast, it is also important to brake strong enough so that a collision can be avoided. As a measure of this aspect, the maximum braking value was examined by conducting a general linear model with the factor warning type. It showed that the maximum braking value varies significantly with the warning type ($F_{(4,55)} = 3.90$, $p = .007$). As can be seen in Fig. 3, except for the RS warning, all driver warnings led to a higher maximum braking value than no warning as in the control group, reaching significance in post-hoc tests (C-RG: $p = .012$; C-RS: $p = .694$; C-AG: $p = .043$; C-AS: $p = .026$). The RS maximum braking value was also significantly lower than in the other three warning groups (RS-RG: $p = .004$; RS-AG: $p = .017$; RS-AS: $p = .010$).

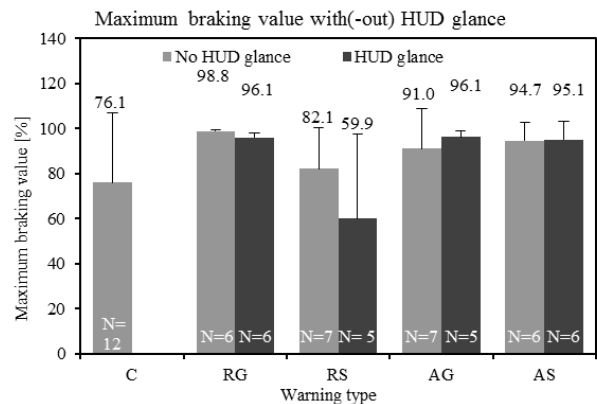


Fig. 3. Mean maximum braking value for the control and all four warning groups with and without head-up display (HUD) glance, displaying the number of participants (N) included in the bottom of the bars.

Moreover, according to another t-test over all warning groups there was no difference in the maximum braking value due to whether the drivers gazed at the HUD or not ($t_{(46)} = 0.67$, $p = .506$; without HUD glance: $M = 91.25\%$, $SD = 14.51\%$; with HUD glance: $M = 87.60\%$, $SD = 22.89\%$).

IV. DISCUSSION

This paper presents an explorative analysis on how different types of driver warnings in the HUD affect drivers' performance (driving and gaze behavior) in a very critical pedestrian crossing scenario. The aim is to create driver warnings which most drivers recognize, understand easily and process rapidly in order to reduce the accident rate and severity in urban areas. An immediate emergency brake or swerving onto the green area would have been ideal as a driver response in the examined scenario. However, all but one driver (not even receiving the RS warning) reacted by braking, since the swerving sign might have not been familiar to the drivers.

Before discussing the results, some limitations have to be mentioned. First of all, the drivers knew that critical situations might occur through instructions and after encountering the first scenario themselves. Thereby drivers were more alert than perhaps in everyday traffic, to which the laboratory environment additionally contributed [15]. This reduced the effect of surprise, which drivers would experience in the field. However, the diversity of the altogether eight critical scenarios included in this study (different safety-critical objects and locations) made it hard to guess what might happen next. Still drivers knew they were being supported by a driver warning system and that the study aimed at evaluating it, so they might have been also more alert for warnings in general. This might have influenced the gaze behavior. For example, drivers' glances might have frequented the HUD more often than they normally would and thus already lasting there when the warning was triggered in the first place. In addition, the driver warnings occurred rather frequently, far more than in real traffic. However, all these circumstances were practical experimental necessities. Furthermore, when interpreting the results, it has to be taken into account that the number of analyzed participants (each of the five groups: $N = 12$) was limited in this study due to (many potential participants being excluded from the experiment as a result of) simulation sickness and experiencing many critical scenarios consecutively being quite stressful for participants.

In general, warning drivers visually in very time critical situations might be questionable as the warning visualization might distract drivers, for example in guiding the drivers' glance onto the warning visualization and capturing it for too long to the cost of the actual safety-critical object being disregarded. However, in the analyzed scenario all drivers gazed at the safety-critical object regardless of whether they had received a warning or not. Moreover, the glance reaction time and the glance duration on the safety-critical object did not differ significantly. Thus, it seems that the examined driver warnings did not distract the drivers from recognizing the safety-critical object, which is also substantiated by the fact that most drivers (around 71%) had their critical object glance before their HUD glance.

With regard to the head-up display, in total 45.8% of the warned drivers showed a HUD glance before beginning to brake ($N = 22$). The glance reaction times were quite short (overall about 0.2 s) indicating that the driver warnings had probably automatically attracted the drivers' attention. Additionally, the glances on the HUD also lasted quite short

supporting the idea that the examined warning visualizations were well-known as it did not take drivers extensively long to process or understand them. However, an exception was the reaction specific warning visualization, which was supposed to elicit a steering reaction from drivers. The glances at the HUD in this group lasted almost double as long as in the other three warning groups. It seems that this sign was less familiar or understandable than the other warning visualizations. It was also not very effective in general as it did not induce any driver to a steering reaction. Moreover, the RS warning type had the weakest effect on the brake reaction time and the maximum braking value as compared to the other warning groups. Thus, the use of well-known and easily understandable warning visualizations seems advisable for such driver warnings.

Since driver warnings aim at safer driving, the driving behavior was of special interest. While the warnings used in this study were not effective in significantly preventing accidents in the examined scenario, there were positive effects of the driver warnings with regard to the brake reaction time and maximum braking value. All driver warnings led to faster and stronger brake reactions. The fact that independent of whether drivers received a warning, still about half of the drivers encountered an accident (in total $N = 31$ collisions) also proves the high criticality of the scenario. As the driver warnings neither increased nor reduced the number of collisions, one might think that the driver warnings presented in the HUD were not seen by the drivers, but the gaze behavior data described above proved the opposite. Drivers did show HUD glances.

However, when drivers gazed at the HUD before brake initiation the positive effect of the driver warnings with regard to the brake reaction time seemed to be evened out as these drivers reached brake reaction times quite similar to the control group. In contrast, the positive effect of a stronger brake reaction induced by the driver warnings was independent of whether drivers showed a HUD glance or not. This difference between the two parameters might be due to the drivers, who gazed at the HUD after brake initiation and before reaching their maximum braking value ($N = 8$), being disregarded. Their inclusion might change the effects again and should thus be further analyzed.

Yet regarding the negated benefit in the brake reaction time, one might assume that visual driver warnings in the HUD are more effective if they are perceived in the peripheral visual field, without directly attracting the drivers' attention or glances, but rather enable them to focus on the outside world and its dangers. This might be an interesting approach for future studies on visual driver warnings. Furthermore, it might explain why the content or the different visualizations of the warning (with exception of the RS warning) did not seem to be crucial. Thus, warning drivers at all appears of importance, less what is displayed in the HUD, as long as it is well-known and easily understood by the drivers. As already discussed above, driver warnings with unfamiliar or complex visualizations might even lead to negative effects or at least negate the potential benefits.

Concluding, in general the driver warnings had positive effects on the drivers' performance. Even though the number

of collisions could not be reduced significantly by the driver warnings, probably also due to the limited time left to act at all, the collision severity was reduced due to faster and stronger brake reactions of the drivers, as intended by the driver warnings in this very critical situation. Only the RS warning is an exception, as it rather diminished the drivers' brake force, even significantly compared to the other warning types. Especially if the RS warning was gazed at, the benefit of receiving a warning was almost evened out in all parameters (number of collisions, brake reaction time and maximum braking value). Thus, the RS warning visualization cannot be recommended for such driver warnings in very critical situations. However, it shows that the gaze behavior should be further investigated when designing visual driver warnings, for its analysis also allowed to mark the other three warning visualizations (RG, AG and AS) as rather supportive than distractive as their HUD glances did not last extensively long (especially in contrast to the RS warning). Thus, the RG, AG and AS warning can be recommended, since they all achieved the driver warnings' aim, while differing little from each other in their effect on drivers' performance.

However, the effect of the warning type still has to be analyzed further in the other seven examined scenarios (following in another paper) as so far it is still unclear, what it is about the examined driver warnings that makes them effective, for example their pure presence, visualization (traffic signs), location (HUD), modality (visual), a combination of those or something else. Further reaction time improvements could be expected by combining the driver warnings (in the HUD) with an additional acoustical or haptic signal, in case drivers are distracted, for example by a not driving related task. Moreover, this study concentrated on rather young drivers, but the effects might change if older drivers are tested. Thus, the driver warnings should also be examined with a group of older drivers since with age the information processing (e.g., [16] [17]) and drivers' performance change and possibly decrease (e.g., [18] [19]) so that older drivers might react differently upon the warning types than younger drivers.

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